Statement of Purpose – Amazon ML Summer School 2026

My journey into AI and Machine Learning began with curiosity about how intelligent systems make decisions and gradually evolved into building practical applications that solve real-world problems. As a Computer Science undergraduate at KIET Group of Institutions, I have actively explored machine learning, deep learning, computer vision, natural language processing, and generative AI through coursework, certifications, and hands-on projects.

One of my most significant projects was a Retrieval-Augmented Generation (RAG) based Question Answering System built using Flutter documentation. I implemented document preprocessing, text chunking, embeddings using all-MiniLM-L6-v2, vector search with ChromaDB, and response generation through an open-source large language model. Building this system helped me understand the complete lifecycle of modern AI applications, from data ingestion and retrieval to inference and evaluation.

I have also worked on NutriLens, a food recognition system developed using Convolutional Neural Networks and the Food-101 dataset. Through this project, I gained practical exposure to image preprocessing, model training, evaluation metrics, and challenges such as overfitting and dataset imbalance. Additionally, I developed an AI Interview Coach that combines NLP techniques, sentiment analysis, keyword relevance scoring, and conversational feedback to help users improve interview performance.

To strengthen my theoretical foundations, I completed certifications including AWS Certified Machine Learning Engineer – Associate, AWS Certified AI Practitioner, AWS Certified Solutions Architect – Associate, and AWS Certified Cloud Practitioner. These experiences exposed me to scalable ML systems, model deployment concepts, MLOps fundamentals, and cloud-based AI workflows.

While I have built several applications, I recognize that my understanding of advanced machine learning theory is still developing. I want to deepen my knowledge of optimization techniques, probabilistic modeling, representation learning, transformers, reinforcement learning, and the mathematical foundations that drive state-of-the-art models. I also wish to learn how researchers analyze, improve, and scale ML systems beyond prototype-level implementations. Amazon ML Summer School provides a unique opportunity to bridge this gap by learning directly from experts who work on cutting-edge machine learning problems.

What makes me a strong fit for this program is not only my technical background but also my willingness to learn and apply knowledge independently. I continuously explore new domains through projects, research-oriented learning, open-source contributions, and competitive coding. I enjoy moving beyond tutorials, understanding how systems work internally, and experimenting with ideas until they become functional solutions.

I view Amazon ML Summer School as an opportunity to transform from a student who builds ML applications into one who understands the deeper principles behind them. The program will help me strengthen my foundations, learn from leading practitioners, and prepare for future contributions in machine learning research and large-scale AI systems.